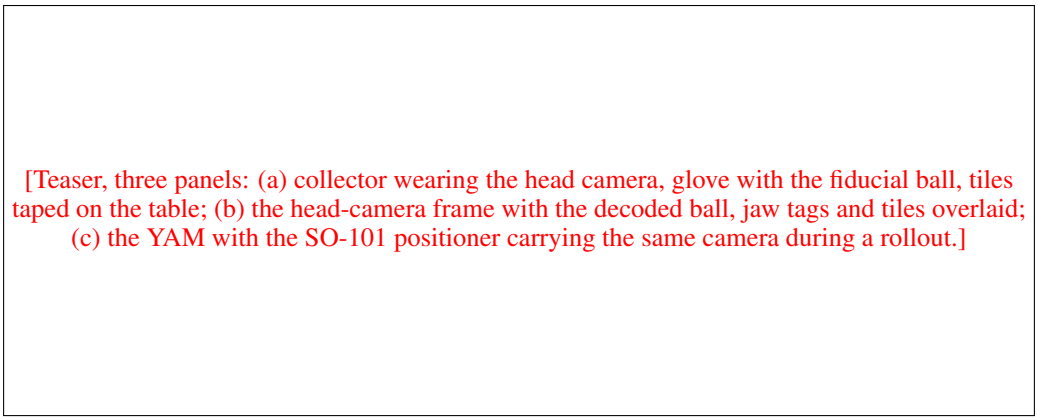


Figure 1: One camera, two mounts. During collection it is on the collector's head and labels the glove from its own view through the tiles; on the robot the same camera rides a six-joint positioner that follows the gripper. Teleoperated data is recorded through the same positioner, so the three corpora (glove, teleop, rollout) share one viewpoint distribution.

One Camera, Worn then Carried: Head-Camera Labelling of Glove Demonstrations and an Arm-Mounted Twin on the Robot

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Abstract: A UMI-style glove needs a label for every frame and a camera for the policy. We use one camera for both, and move it with the operator: a 4K camera on the collector's head sees the fiducial ball on the glove and the printed tiles taped around the workspace, so every frame carries the gripper's pose in a world frame fixed by the tiles and the jaw width from the same view, with no fixed infrastructure, no IMU and no SLAM. On the robot a six-joint positioner carries the same camera and keeps the gripper in view the way the head kept the hand in view, so the policy sees at deployment what it saw in training. The labels are checked against the robot's encoders: $6.5\,\mathrm{mm}$ median tool-point error, 1.6° rotation, and 93–$100\,\%$ of frames labelled per session, against 78–$92\,\%$ for a fixed external camera that the collector's body hides. One person records 2.0–3.1 demonstrations per minute with the glove against 1.6 by teleoperation. Adding 194 glove segments to 172 teleoperated episodes lowers ACT's loss on held-out teleoperated episodes from 0.318 to 0.263; added to 48 teleoperated episodes it reaches 0.295, the level of 96–172 teleoperated episodes alone, and on the robot the 48-episode co-trained policy succeeds in X/25 trials against X/25 for 48 teleoperated episodes alone and X/25 for the full 172. Against the conventional wrist-camera and fixed-camera recipes recorded from the same demonstrations the head-camera policy succeeds in X/25 vs X/25 and X/25. [final numbers from the grid]

1 The idea

Our earlier glove [1] was tracked by a fixed 4K camera over the workspace, which gives absolute labels but ties collection to a tripod and loses the ball whenever the collector's body comes between camera and hand. The workshop's question is what a UMI should look like when it is cheap, deployable and gives the policy the right observation. Our answer is to let the label camera be worn: the collector's head follows the hand by habit, keeps the gripper roughly centred and roughly at arm's length, and is never occluded by the collector's own torso. The world frame that a fixed camera provided for free is replaced by printed AprilTag tiles taped on the table, which cost nothing, survive being nudged, and are solved in every frame that shows one. The same camera then becomes the policy's observation. On the robot a small six-joint arm (an SO-101) carries it and imitates the head: it keeps the gripper where the collector's head kept the hand, from the robot's joint state alone, identically during teleoperation and during rollouts. Nothing in the pipeline depends on a fixed camera; one was kept only as an ablation.

2 System

Collection. The glove is the fiducial-tracked YAM gripper replica of Shin [1]: the robot's own fingers on a rack-and-pinion linear mechanism, an 11-face $15\,\mathrm{mm}$ ArUco dodecahedron on the back of the hand, two AprilTags in the jaws. The collector wears an Arducam B0587 4K camera [mount: how it is worn, weight, focus setting] at $15\,\mathrm{Hz}$, exposure locked short enough to freeze hand motion. Four $60\,\mathrm{mm}$ AprilTag tiles are taped around the workspace (Fig. 1); their relative layout is measured once from the head-camera footage itself (§2). Cameras roll continuously and a key press marks episode boundaries; the collector records 2.0–3.1 pick-and-place demonstrations per minute (median episode 6–8 s), against 1.6 per minute by teleoperation of the same task on the same arm (median $14\,\mathrm{s}$).

Labels from the head camera. Each frame yields three things from one image. (i) The camera's pose in the tile frame, from every tile in view; (ii) the ball's pose in the camera, by fused PnP over the visible faces; (iii) the jaw width, from the jaw tags. (i) and (ii) compose to the gripper's pose in the tile frame, and a tile-to-robot-base link, fitted once from a session in which the robot carried the ball, maps it into the robot's base frame for the same joint-space IK the teleoperated labels use. Two things make (i) robust on a moving camera. First, the tile layout is refit from the footage: the tile positions declared from the first fixed-camera session were inconsistent at the few-millimetre level, so solves over three or four tiles ran at $3.4\,\mathrm{px}$ RMS while any pair ran under $1\,\mathrm{px}$; refitting the layout from single-tile poses observed in one session brings the four-tile solve to $0.5\,\mathrm{px}$ and the disagreement between two single-tile camera poses on the same frame from 10–$13\,\mathrm{mm}$ to 4–$9\,\mathrm{mm}$ median. Second, frames that show only one tile are chained to the last accepted pose under a motion bound ($0.6\,\mathrm{m/s}$, $120\,^\circ/\mathrm{s}$) instead of being discarded. What remains blind is cut: stretches longer than $0.3\,\mathrm{s}$ without a label split an episode into segments, and segments under $1\,\mathrm{s}$ are dropped, so a blind moment costs its frames and not the episode. Per session, 0.7–$7.4\,\%$ of frames are cut this way; the same demonstrations labelled from the fixed external camera lose 8–$22\,\%$, because the collector's body hides the ball, and fragment into 259 segments where the head camera keeps 200.

The robot's head. A six-joint SO-101 arm beside the YAM carries the same camera (Fig. 1c). At the start of a session it parks at the median head pose of the glove sessions, closed-loop against the tiles, and calibrates its own view axes against them. During teleoperation and during rollouts it reads only the YAM's joint state: forward kinematics gives the gripper's position, a fixed gain keeps the gripper's bearing in the camera where it was at park, and a residual model fitted on the glove sessions reproduces how the head's viewpoint depends on the hand's position. Commands pass through a low-pass filter, a rate cap of $50\,^\circ/\mathrm{s}$ and a workspace box, and the resulting camera motion lags the gripper by about $0.4\,\mathrm{s}$, with a median camera position error of $16\,\mathrm{mm}$ ($34\,\mathrm{mm}$ p90) against the tiles. The gripper's framing in the robot's head-camera footage (centroid at 0.37/0.62 of the image width/height) matches the collector's (0.38/0.60). The kinematic model of the arm is fitted

Table 1: The eight policies. Teleop labels are encoders except arm 6. Robot val = ACT loss on the 6 held-out teleoperated episodes (encoder targets, comparable across all rows except 6). Rollouts: 25 placements, one trial each, 60 s budget; failure modes n = no grasp, d = dropped before the bin, b = dropped at the bin. [fill from the grid]

#	arm	glove labels	observation	teleop + glove	val	success	fails
1	teleop only	—	head cam	172	0.318	/25	
2	**device**	head cam	head cam	172 + 194	**0.263**	/25	
3	teleop only, low budget	—	head cam	48	0.464	/25	
4	device, low budget	head cam	head cam	48 + 194	0.295	/25	
5	wrist-camera UMI	fixed cam	wrist fisheye	168 + 253	0.257	/25	
6	head-cam teleop labels	—	head cam	172 (head labels)	(0.282)	/25	
7	fixed camera as system	fixed cam	fixed cam	172 + 255	0.256	/25	
8	head labels, wrist view	head cam	wrist fisheye	168 + 192	0.252	/25	

once from 60 tile-verified poses (7 mm/0.7° residual) and the camera is released to the recorder after calibration; the positioner is otherwise invisible to the policy.

Datasets. Teleoperation under the positioner gave 178 episodes (38.6k frames at 15 Hz); 6 are held out as the validation set for every arm. Four glove sessions gave 198 demonstrations, 200 segments after blind cutting (50.0k frames), 6 held out. Every policy is ACT [2] trained with LeRobot [3]: one recipe, 50k steps, joint-space actions, best checkpoint by validation loss on the robot's held-out episodes. The glove timeline is resampled to teleoperation pace and the measured width is mapped to the robot's commanded value, as in Shin [1].

3 Experiments

Design. In every arm the teleoperated half carries the robot's encoder labels; only the glove half carries the camera under test, and its observation is whichever camera the arm deploys. Table 1 lists the eight arms. Arms 1–4 are the device and its budget pair; arm 6 checks the labels; arms 5, 7 and 8 compare cameras. All glove arms train on the same 198 demonstrations. The fixed-camera arms train on 10 % fewer glove frames (43.5k against 48.5k), which is the occlusion cost of the fixed camera and not a budget choice.

Label accuracy. With the ball on the YAM and the encoders as reference, the head-camera chain places the tool point with 6.5 mm median / 14.1 mm p90 error and 1.55° / 2.70° rotation (5 065 frames, 91 % inliers), after the one-time base link absorbs the constant offset. Downstream, a policy trained on head-camera labels instead of encoders (arm 6 against arm 1) [scores X on encoder targets, Y% above the encoder-trained twin; sweep running] and succeeds in X/25 rollouts against X/25.

Co-training and the teleoperation budget. Adding the glove set to all 172 teleoperated episodes lowers the robot-validation loss from 0.318 to 0.263 (17 %); added to 48 teleoperated episodes it lowers it from 0.464 to 0.295. Fig. 2 shows the full curve: the glove set is worth 65–100 teleoperated episodes by log-linear interpolation of the teleop-only curve, i.e. 2–3 glove demonstrations per teleoperated one, and the 48-episode mix lands between 96 and 172 teleoperated episodes alone. The gain shrinks with budget (0.075 at 24 episodes down to 0.011 at 96, measured against the full mix), so glove data stretches a small teleoperated set but does not replace the last few dozen episodes. On the robot the 48-episode mix succeeds in X/25 against X/25 for the 48 teleoperated episodes alone and X/25 for the full 172, [with the recovery behaviour of the mix: in scene 5 the mix missed its first grasp, backed off and re-grasped, where the teleop-only policy never closed in].

Which camera. Offline, the four co-trained arms with different cameras (2, 5, 7, 8) sit within 0.011 of each other, and each gains 0.04–0.06 over its teleop-only twin, so the validation loss does not separate the cameras. The rollouts do. Arm 2 succeeds in X/25, the wrist-camera recipe in X/25,

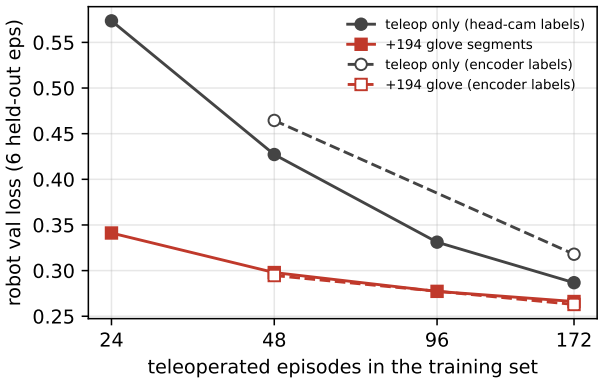


Figure 2: Robot validation loss against the number of teleoperated episodes, with and without the 194 glove segments. Solid: head-camera-labelled teleoperation (round-2 chain, four budgets); dashed: encoder-labelled teleoperation (final design, two budgets). The glove set moves every budget to the level of 2–4× as many teleoperated episodes.

[Rollout protocol figure as in the device paper: the 25 placements with per-scene success over the 8 arms; ghost-alignment inset. Generate from rollout_logs/grid_20261016 (placements.json, scene*_ref.png, trials.jsonl).]

Figure 3: Rollout protocol. 25 placements, each scene's block re-placed to a saved reference for every arm (ghost alignment), arm order shuffled per scene, 60 s budget.

the fixed camera in X/25, and head labels on the wrist view in X/25. The failure modes split by camera: head-view policies fail by never grasping, wrist-view policies by dropping the block during the carry, d+b = X of Y failures for arms 5 and 8 against X of Y for arms 1–4. [every observed wrist-view drop happened while the bin was outside the fisheye's view; no drop was observed with the bin in view; the head camera keeps gripper and bin in frame throughout the carry.] Holding the labels fixed (arm 8 against arm 2) isolates the observation; holding the observation fixed (arm 8 against arm 5) isolates the label source: result. The fixed camera (arm 7) result; note that it was mounted low because the positioner occupied the elevated vantage point, so its occlusion rate is an upper bound for a fixed camera in this workspace.

Anecdotes. [Transcribe from rollout_logs/grid_20261016/anecdotes.md: scene-5 budget pair; mix_gHA opening mid-carry with the block out of view; wrist drops only when the bin is out of view.]

4 Limitations and what is next

The tile layout is refit from one session by median single-tile poses rather than bundle-adjusted; a proper adjustment should bring the single-tile jitter below the current 4–9 mm. The positioner lags the gripper by 0.4 s and hits joint limits on the largest sweeps; a lighter camera or a faster arm would close that. The head camera also sees the collector's arm and the glove's fingers, which is the appearance gap that keeps glove-only policies from transferring (robot-validation loss 0.52

against 0.26 co-trained), as in the fixed-camera version of the device. The fixed-camera ablation was mounted low because the positioner took the high vantage point; a higher fixed camera would occlude less, though it cannot follow the hand. Hardware and software are open [1, 4]. [what is next: multi-tile workspaces, in-the-wild sessions, a second task]

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